NOVORA: Technical Architecture & ML Summary

End-to-End System Specifications • Frontend to Backend • Multi-Model Machine Learning Training

1. FULL TECHNOLOGY STACK (END-TO-END)

Layer	Technologies	Role & Core Responsibility
Frontend (UI/UX)	HTML5, Tailwind CSS, Chart.js, FontAwesome	Reactive single-page dashboard: direct numeric inputs, dynamic Chart.js cohort scatter plot with active student highlight, grade distribution histogram, and persistent localStorage history log.
Backend API	Node.js Serverless Functions (`api/predict.js`)	Zero-cold-start REST endpoint with CORS support. Validates inputs, calculates blended inference, value-add trajectory, and target gap analysis.
Machine Learning	Python 3.13, Scikit-Learn, Pandas, NumPy, SciPy	Data ingestion, 80/20 train-test partitioning, zero-leakage StandardScaler scaling, 6-algorithm benchmark, and .joblib model serialization.
Deployment	Vercel (`vercel.json`), Git, GitHub	One-click automated cloud deployment with serverless routing rules, rewrite paths, and semantic per-phase Git version history.

2. DATASET SPECIFICATIONS & INPUT FEATURES

Dataset contains **400 verified student records** (100% complete, 0 missing nulls, 0 duplicate instances). The objective is to predict final summative **Exam_Score (0 - 100)** using behavioral and formative indicators.

Feature Name	Data Type	Observed Range	Correlation (r)	Educational Significance
Study_Hours	float64	1.05 – 9.91 hrs	**+0.8088**	Weekly independent study hours; strongest linear predictor.
Attendance	float64	50.5% – 99.9%	+0.3161	Classroom presence rate; ensures instructional exposure.
Assignments	int64	0 – 10 tasks	+0.4229	Formative coursework submissions; active problem solving.
Marks_Obtained	float64	0.0% – 100.0%	Integrated Anchor	Midterm / internal continuous evaluation baseline for trajectory analysis.
Exam_Score (Target)	float64	24.38 – 100.0%	**1.0000**	Final comprehensive examination outcome (Mean: 67.35, Std: 16.33).

3. MACHINE LEARNING MODEL TRAINING & LEADERBOARD

Partitioning Strategy: 80% Training (320 samples) / 20% Held-Out Testing (80 samples) with fixed random_state=42. Strict 5-Fold Cross-Validation was performed across all candidate algorithms to ensure zero overfitting.

Rank & Model Name	Test R^2	Adj. R^2	MAE (Marks)	RMSE	5-Fold CV R^2	Outcome Status
■ **Linear Regression (OLS)**	**0.9064**	0.9027	**4.0298**	5.0432	0.8984 ± 0.015	**Champion Model**
■ **Ridge Regression (L2)**	0.9064	0.9027	4.0306	5.0436	0.8984 ± 0.015	Regularized Linear
■ **Lasso Regression (L1)**	0.9062	0.9025	4.0390	5.0500	0.8984 ± 0.015	Sparse Baseline
4. Gradient Boosting	0.8663	0.8610	4.7634	6.0291	0.8681 ± 0.012	Ensemble Boosting
5. Random Forest (100 Trees)	0.8499	0.8440	5.0829	6.3867	0.8532 ± 0.016	Ensemble Bagging
6. Decision Tree Regressor	0.7758	0.7669	6.1485	7.8065	0.7804 ± 0.040	Single Tree Baseline

Why OLS Linear Regression Won: The dataset exhibits strong linear determinism (r = 0.81 with study hours) with near-zero inter-predictor collinearity (|r| < 0.12). OLS provided the lowest prediction error (MAE: ±4.03 marks) with optimal coefficient interpretability.

4. MATHEMATICAL FORMULATION & TRAJECTORY ENGINE

Component	Exact Mathematical Equation & Weight Allocation	Pedagogical Interpretation
Behavioral Base Score	`Base_Score = 3.4209 + 4.7639*(Study) + 0.3795*(Attend) + 1.9669*(Assign)`	+1 hr study = **+4.76 marks** +10% attendance = **+3.80 marks** +1 assignment = **+1.97 marks**

Blended Final Prediction	`Final_Score = 0.40*(Marks_Obtained) + 0.60*(Base_Score)`	Combines prior formative midterm baseline (40%) with current study habit velocity (60%).
Value-Add Trajectory (Δ)	`Δ = Final_Score - Marks_Obtained`	Δ > +3.0: **Accelerating Growth** -3.0 ≤ Δ ≤ +3.0: **Steady Trajectory** Δ < -3.0: **Declining Warning Alert**
Target Gap Analysis	`Hours_Needed = (Target_Score - Final_Score) / (4.7639 * 0.60)`	Quantifies exact study hours or homework tasks required to elevate to next tier (Distinction 80%).

5. END-TO-END WORKFLOW & APPLICATION MODULES

Workflow: 1. Data Input: Educator or student inputs Marks Obtained, Study Hours, Attendance, and Assignments. **2. Real-Time Inference:** Client-side JS computes instant score (with serverless /api/predict API fallback). **3. Diagnostics:** Calculates Value-Add Trajectory (Growth/Decline), Risk Tier (High Performer, Moderate, At Risk), and Gap to Distinction. **4. Persistence:** Prediction logged to browser localStorage with search, filtering, and CSV export. **5. Action Plan:** System outputs a dynamic 4-Week Intervention Roadmap with prioritized weekly milestone goals.

Deployment: Vercel Cloud (Zero-Config via vercel.json) | **GitHub Repository:** https://github.com/Abhi111227/Student-Performance-Prediction-Framework | **Champion Model:** Ordinary Least Squares (R^2 = 90.64%)